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OpthaEdge-DED: An Offline Edge-AI System for Diabetic Retinopathy Grading Using Ordinal-Preserving Knowledge Distillation from a Swin Transformer to a Compact MobileNetV3

Submitted in partial fulfillment of the requirements of the degree of

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by

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Abstract

Diabetic retinopathy (DR) is a leading cause of preventable blindness, but screening depends on specialists and equipment that are scarce in rural and low-resource settings. Deep models that grade DR accurately are large and need GPU servers, while portable devices need models that run offline on low-power hardware. This project presents OpthaEdge-DED, an offline system that combines a low-cost imaging assembly (a Raspberry Pi 4, a 20D lens and a stability stand) with a compact deep model, and a web platform that places the model in a clinical screening workflow.

A Swin-Base transformer with a CBAM attention module (87.5 M parameters, 384 × 384 input) was trained on APTOS 2019 as a teacher, with a two-stage retinal preprocessing pipeline, focal loss and a weighted sampler. Its knowledge was distilled into a MobileNetV3-Large student with a lightweight CBAM (3.05 M parameters, 224 × 224 input) using temperature-scaled KL divergence, cross-entropy and feature mimicking. The Ordinal Consistency Score (OCS) was proposed to test whether the ordered structure of the DR grades survives distillation. The student was exported to ONNX and quantised to INT8.

On a held-out stratified validation split (733 images), the teacher reached an accuracy of 82.13% and a quadratic weighted kappa (QWK) of 0.9043. The student, with 28.7 times fewer parameters, reached 81.45% and 0.9033, with a macro AUC of 0.942. The OCS was 0.7954 for the teacher and 0.7950 for the student, so the ordinal structure was transferred. About 82% of the student's errors were between neighbouring grades, and as a screening test it detected 99.2% of DR cases and 91.9% of referable cases. The ONNX file shrinks from 373 MB (teacher) to 12.2 MB (FP32) and 3.55 MB (INT8); INT8 lowers accuracy to 78.4% and QWK to 0.890. Measurements on the Raspberry Pi, a test on external data and clinical validation are still to be done.

Keywords: diabetic retinopathy, knowledge distillation, Swin Transformer, MobileNetV3, CBAM, ordinal classification, quadratic weighted kappa, edge AI, ONNX, INT8 quantisation, Raspberry Pi.

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Abbreviations

APTOS	Asia Pacific Tele-Ophthalmology Society
AUC	Area Under the (ROC) Curve
CBAM	Convolutional Block Attention Module
CE	Cross-Entropy
CLAHE	Contrast-Limited Adaptive Histogram Equalisation
CNN	Convolutional Neural Network
DR	Diabetic Retinopathy
FP32	32-bit Floating Point
INT8	8-bit Integer
KD	Knowledge Distillation
KL	Kullback–Leibler (divergence)
MSE	Mean Squared Error
OCS	Ordinal Consistency Score
ONNX	Open Neural Network Exchange
PTQ	Post-Training Quantisation
QDQ	Quantise–Dequantise (ONNX format)
QWK	Quadratic Weighted Kappa
RBAC	Role-Based Access Control
ROI	Region of Interest
ROC	Receiver Operating Characteristic
ViT	Vision Transformer

Chapter 1

Introduction

1.1 Background

Vision allows people to read, work, travel and live independently, yet it is increasingly threatened by vascular disease of the eye. The commonest of these conditions, diabetic retinopathy (DR), is among the leading causes of blindness in people with diabetes. DR is a microvascular complication in which chronic hyperglycaemia damages the small vessels of the retina, the light-sensitive lining at the back of the eye. The damaged vessels produce microaneurysms, haemorrhages and exudates and, as the disease advances, can lead to irreversible visual impairment.

More than 537 million adults live with diabetes worldwide, and roughly 35% of them develop some form of DR, an estimated 93–103 million people. This number is expected to rise to about 161 million by 2045. Because DR is usually asymptomatic in its early stages it is often called a “silent disease”, and by the time symptoms appear the damage may already be permanent. Early detection and timely referral are therefore the most effective ways to prevent vision loss.

1.2 Motivation

The reference standard for retinal examination, a tabletop fundus camera or optical coherence tomography (OCT), is large, stationary and expensive, often costing more than USD 10,000. Mass screening with such devices is impractical in remote and low-income regions where medical facilities are scarce. Manual grading of retinal photographs is also repetitive and subjective, and trained graders are in short supply: in India, for example, roughly 15,000 eye-care specialists serve a diabetic population of about 70 million.

Low-cost fundus imaging with an indirect-ophthalmoscopy lens, combined with deep learning, offers a way to move screening to the point of care. Two problems stand in the way. First, the most accurate models, such as vision transformers, are too large for low-power edge hardware. Second, imaging prototypes and deployable grading software are usually developed separately, so there are few complete, offline systems that can both capture an image and grade its DR severity.

1.3 Proposed Work

This project presents OpthaEdge-DED, a hardware–software screening system. On the hardware side, a Raspberry Pi 4 Model B, a 20D fundus lens and a focus-oriented portable stand capture retinal images in a reproducible way. On the software side, a high-capacity teacher network (a Swin-Base transformer with a CBAM attention module, 87.5 million parameters) is trained on a GPU server, and its knowledge is distilled into a lightweight student (MobileNetV3-Large with a lightweight CBAM, 3.05 million parameters). The student is exported to ONNX and quantised to INT8 so that it can run offline on the Raspberry Pi 4. DR is graded on the five-point international scale: 0 (no DR), 1 (mild), 2 (moderate), 3 (severe) and 4 (proliferative).

Because DR grades are ordered, we also define the Ordinal Consistency Score (OCS), a simple measure of whether a model's probability vector falls steadily as the grade moves away from the predicted one. OCS is used to check whether ordinal structure survives the transfer from teacher to student.

1.4 Contributions

1. **A portable, offline screening system** built from a Raspberry Pi 4, a 20D lens and a stability stand.
2. **A teacher–student framework** in which a Swin-Base + CBAM teacher is distilled into a MobileNetV3-Large + lightweight-CBAM student using temperature-scaled KL divergence, hard-label cross-entropy and feature mimicking. On the held-out validation split the student reaches a quadratic weighted kappa (QWK) of 0.903, against 0.904 for the teacher, with 28.7 times fewer parameters.
3. **The Ordinal Consistency Score (OCS)**, a temperature-invariant metric for the ordinality of predicted class probabilities.
4. **A reproducible deployment pipeline**: ONNX export with the normalisation folded into the graph, and INT8 post-training quantisation calibrated on 200 training images.

1.5 Organisation of the Report

Chapter 2 reviews the related literature and identifies research gaps. Chapter 3 states the problem and the objectives. Chapter 4 describes the methodology: hardware, data, preprocessing, the teacher and student networks, distillation, OCS and deployment. Chapter 5 reports the results obtained so far and discusses them. Chapter 6 concludes the report and outlines future work. The references and the acknowledgement follow Chapter 6.

Chapter 2

Literature Study

2.1 Literature Survey

2.1.1 Portable and smartphone-based fundus imaging

Several groups have built low-cost retinal imagers by pairing a camera with a 20D lens. Kaviya *et al.* [23] used a smartphone and a PVC-tube adapter with deep models (ResNet50, MobileNetV2, U-Net) and Gaussian-filter and CLAHE pre-processing, reporting more than 90% accuracy for glaucoma and DR detection. Thaware *et al.* [24] built a Raspberry Pi imager and deployed an EfficientNet-B0 model with spatial attention, trained on about 13,300 images, reporting 96.2% accuracy and about 5 s on-device inference. Iqbal [25] and Wintergerst *et al.* [26] reviewed smartphone fundus photography and concluded that image-quality variability and the lack of standard protocols are the main obstacles. Mittal *et al.* [27] piloted a D-EYE smartphone ophthalmoscope in 110 patients and found that video capture was better than still images. Senthil *et al.* [28] evaluated an offline glaucoma AI running on a smartphone camera in 213 adults, with 92.02% accuracy, and noted weaker performance in early disease.

2.1.2 Deep learning for DR grading

Convolutional networks and, more recently, vision transformers achieve strong results on public DR datasets such as APTOS 2019 [1] and the EyePACS/Kaggle competition data [2]. The Swin Transformer [3] is attractive for fundus images because its shifted-window attention captures both local lesion texture and global retinal structure at high resolution without the quadratic cost of full self-attention. Attention modules such as CBAM [4] further direct the model toward lesion-dense regions, and self-supervised pre-training has also been explored for DR grading [22]. These models are, however, designed for GPU servers.

2.1.3 Knowledge distillation for DR

Knowledge distillation [10] trains a small student to match the softened outputs of a large teacher. In DR, Luo *et al.* [11] introduced self-knowledge distillation with CAM attention and a mimicking module. Moya-Albor *et al.* [12] used InceptionV3 as both teacher and student with KL divergence and categorical cross-entropy. Islam *et al.* [19] distilled a dual-teacher ensemble (Swin Transformer and ResNet152V2) into an Xception student enhanced with CBAM, reporting 99.04% accuracy on APTOS. Rautaray *et al.* [20] distilled a FastViT-MA26 teacher into EfficientNet-B0 for DR severity classification (95.39% accuracy). Park *et al.* [21]

proposed Style-KD for class-imbalanced medical image classification, and Yilmaz and Aiyengar [14] showed cross-architecture ViT-to-CNN distillation on a GPU-equipped Jetson Nano.

2.1.4 Ordinal learning

DR grades form an ordered scale, and Cohen's quadratic weighted kappa (QWK) [5] is the standard agreement measure because it penalises errors by the squared grade distance. Díaz and Marathe [13] proposed soft ordinal label encoding (SORD), which builds soft targets from ground-truth label distances. Focal loss [6] focuses learning on hard, minority-class samples, and a differentiable QWK term can be combined with it to make the loss ordinal-aware.

2.2 Literature Review Table

Tables 2.1 and 2.2 summarise the portable-imaging and knowledge-distillation studies reviewed above.

Table 2.1. Summary of representative portable-imaging and edge-AI studies.

#	Study	Approach	Data / result	Limitation noted
1	Kaviya *et al.* [23] (2025)	Smartphone + 20D lens; ResNet50, MobileNetV2, U-Net	DRIVE, STARE, DIARETDB1; >90% accuracy (glaucoma, DR)	Binary detection; no severity grading
2	Thaware *et al.* [24] (2025)	Raspberry Pi imager; EfficientNet-B0 + spatial attention	~13,300 images; 96.2% accuracy, ~5 s on-device	Compute-heavy for edge; limited explainability
3	Iqbal [25] (2021)	Narrative review of smartphone fundus photography	Review	Need for standard protocols and clinical validation
4	Wintergerst *et al.* [26] (2020)	Review of smartphone vs. conventional imaging	Review	Image-quality variability; no standard grading
5	Mittal *et al.* [27] (2025)	D-EYE smartphone ophthalmoscope pilot	110 patients; video better than stills	Needs larger studies and automated grading
6	Senthil *et al.* [28] (2025)	Offline glaucoma AI on smartphone camera	213 adults; 92.02% accuracy	Weaker in early disease; needs broader validation

Table 2.2. Representative knowledge-distillation approaches for DR.

Study	Teacher → student	Dataset	Reported accuracy	Edge test
Luo *et al.* [11]	Self-distillation with CAM attention	Public DR data	See [11]	No
Moya-Albor *et al.* [12]	InceptionV3 → InceptionV3	Fundus lesions	See [12]	No
Islam *et al.* [19]	Swin + ResNet152V2 → Xception + CBAM	APTOS	99.04%	No
Rautaray *et al.* [20]	FastViT-MA26 → EfficientNet-B0	DR severity data	95.39%	No
Yilmaz & Aiyengar [14]	ViT → CNN	Fundus images	See [14]	Jetson Nano

2.3 Research Gaps

The studies above leave the following gaps, which motivate this work:

- **Hardware and software are rarely integrated.** Imaging prototypes and deployable severity graders are mostly developed separately, and severity grading (as opposed to binary detection) is weak on edge devices.
- **Unstable handheld capture.** Smartphone adapters suffer from focus instability and depend on examiner skill.
- **No measure of soft-label ordinality.** Accuracy and QWK evaluate hard predictions; none evaluates the shape of the probability vector that distillation actually transfers.
- **Ordinal structure in distillation is unexplored.** The DR distillation methods reviewed train the teacher with cross-entropy, and whether an ordinal-aware teacher gives better soft labels has not been investigated.
- **Limited validation and transparency.** Most models are evaluated on a single dataset, with little external or clinical validation and limited explainability.

Chapter 3

Problem Statement

After studying the literature and identifying the gaps above, we formulated the following problem statement:

> *"To design and develop OpthaEdge-DED, an edge-AI-based portable fundus imaging system that captures retinal images with a 20D lens and automatically grades diabetic retinopathy severity offline and in real time, improving early diagnosis and access to eye care."*

3.1 Objectives

1. To design a portable and mechanically stable fundus imaging setup using a 20D lens and a Raspberry Pi camera.
2. To pre-process fundus images for illumination correction, contrast enhancement and removal of non-retinal borders.
3. To develop a high-accuracy teacher model (Swin-Base + CBAM) for five-grade DR classification.
4. To distil the teacher into a lightweight student (MobileNetV3-Large + lightweight CBAM) suitable for edge hardware.
5. To define and evaluate the Ordinal Consistency Score (OCS) for the predictions made by the distilled model.
6. To export the student to ONNX, quantise it to INT8 and measure its size, accuracy and latency on a Raspberry Pi 4.

3.2 Scope

The present work covers DR grading from fundus photographs. Explainable-AI visualisation (Grad-CAM++), other retinal diseases such as glaucoma, and multimodal data are outside the implemented scope and are listed as future work in Chapter 6.

Chapter 4

Methodology

4.1 Overview

OpthaEdge-DED is a portable, AI-powered system for grading diabetic retinopathy (DR) in resource-constrained settings. It integrates a custom-built fundus imaging assembly with an optimised deep-learning model that runs on a Raspberry Pi 4 Model B, so that no internet connection or specialist supervision is needed. The system has five parts: (1) a hardware imaging assembly, built around a 20D ophthalmic lens on a portable stability stand; (2) an image preprocessing pipeline; (3) a high-capacity teacher network trained on a GPU server; (4) a knowledge-distillation stage that transfers the teacher's knowledge to a lightweight student network; and (5) an INT8-quantised ONNX model for real-time inference on the edge device.

The methodology follows a staged pipeline (Fig. 4.3): data acquisition and preprocessing, teacher training (Phase A), knowledge distillation (Phase B), student design (Phase C), and quantisation and deployment (Phase D). The teacher is used only for training. Only the student is deployed.

4.2 Hardware Setup and Imaging Assembly

The hardware addresses a gap identified in Chapter 2: the lack of a stable, affordable and portable fundus imaging platform. It has three physical components that together give consistent, reproducible retinal capture (Fig. 4.1).

4.2.1 Compute Module

The Raspberry Pi 4 Model B is the central compute unit. It has a 1.8 GHz quad-core Cortex-A72 processor, 4 GB of LPDDR4 RAM, and USB 3.0 and MIPI CSI-2 camera ports. The board supports both offline inference and real-time image processing, which suits point-of-care use where cloud connectivity cannot be assumed. A 64 GB Class-10 microSD card stores the operating system, the model weights and the captured images.

4.2.2 Fundus Lens and Optical Assembly

A Volk 20D indirect-ophthalmoscopy lens focuses light from the retina. It gives a wide field of view of about 46° and is well suited to non-mydriatic imaging. The lens is mounted in a custom PVC-pipe housing that aligns it with a Raspberry Pi Camera Module v2 (8 MP, Sony

IMX219 sensor). The camera captures images of 3280 × 2464 pixels with a fixed focal length and a manually set aperture chosen for retinal contrast.

4.2.3 Portable Stability Stand

Handheld capture is a key weakness of earlier smartphone-based systems, because it makes image quality depend on the examiner. OpthaEdge-DED therefore uses a purpose-built, focus-oriented portable stand that holds the imaging assembly fixed relative to the patient's eye. The stand has a chin rest, a forehead brace and a motorised fine-focus rail, so the operator can obtain sharp, reproducible images without specialist ophthalmic training. The mechanical design reduces motion blur and removes examiner-dependent variability.

[FIGURE PLACEHOLDER]

Insert a photograph of the assembled prototype with labels: (a) Raspberry Pi 4, (b) camera module, (c) 20D lens in PVC housing, (d) chin rest / forehead brace, (e) fine-focus rail.

Fig. 4.1. The OpthaEdge-DED imaging prototype. [Add the photograph and component labels (a)–(e).]

4.3 Data Collection

4.3.1 Data Sources

Two public datasets are used. The **APTOS 2019 Blindness Detection** dataset [1] contains 3,662 labelled fundus photographs collected in rural screening programmes, graded 0 to 4, and is representative of real screening conditions. All training and evaluation results in Chapter 5 use APTOS. The **EyePACS** dataset [2] contains 35,126 labelled images with the same five grades, but is heavily skewed toward grade 0 (25,810 images). For EyePACS, a fixed balanced pool of 5,000 images (1,000 per grade, with offline augmentation for the two rarest grades) was prepared in a separate notebook, and a preliminary experiment on it is reported in Section 5.6. Images captured with the OpthaEdge-DED prototype are reserved for qualitative testing under deployment conditions.

4.3.2 Dataset Structure and Split Strategy

The APTOS images are divided once into an 80% training subset (2,929 images) and a 20% validation subset (733 images) by stratified sampling with a fixed random seed of 42, so each subset keeps the original class proportions. The split is saved to disk and reloaded in every run, so all experiments use identical data. Table 4.1 gives the class distribution. The data are strongly imbalanced: grade 0 makes up about 49% of the images, and grade 3 only about 5%.

Table 4.1. Class distribution of the APTOS 2019 dataset and the train/validation split.

Subset	Grade 0	Grade 1	Grade 2	Grade 3	Grade 4	Total
All images	1,805	370	999	193	295	3,662
Training (80%)	1,444	296	799	154	236	2,929
Validation (20%)	361	74	200	39	59	733

4.4 Data Preprocessing

4.4.1 Image Preprocessing

Raw fundus photographs vary widely in illumination, contrast and the amount of black border. A deterministic two-stage pipeline, applied identically in training and in deployment, normalises them. **Stage 1** (denoising and local contrast normalisation) subtracts a Gaussian-blurred copy of the image from a weighted copy of itself:

$$I_e(x, y) = \alpha \cdot I(x, y) + \beta \cdot (G_\sigma * I)(x, y) + \gamma \tag{4.1}$$

where G_σ is a Gaussian filter, and $\alpha = 4$, $\beta = -4$, $\gamma = 128$ and $\sigma = 10$. This removes uneven illumination and emphasises fine structures such as vessels, microaneurysms, haemorrhages and exudates. **Stage 2** (region-of-interest selection) thresholds the greyscale image with Otsu's method and crops to the bounding box of the foreground, which removes the black border around the retina. The cropped image is resized to 384 × 384 pixels with area interpolation and saved as a PNG. For the student, the same 384 × 384 image is downsampled to 224 × 224 inside the model, so both networks see the same pre-processed image. Pixel values are finally normalised with the ImageNet mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225). Fig. 4.2 shows one example per DR grade at each stage.

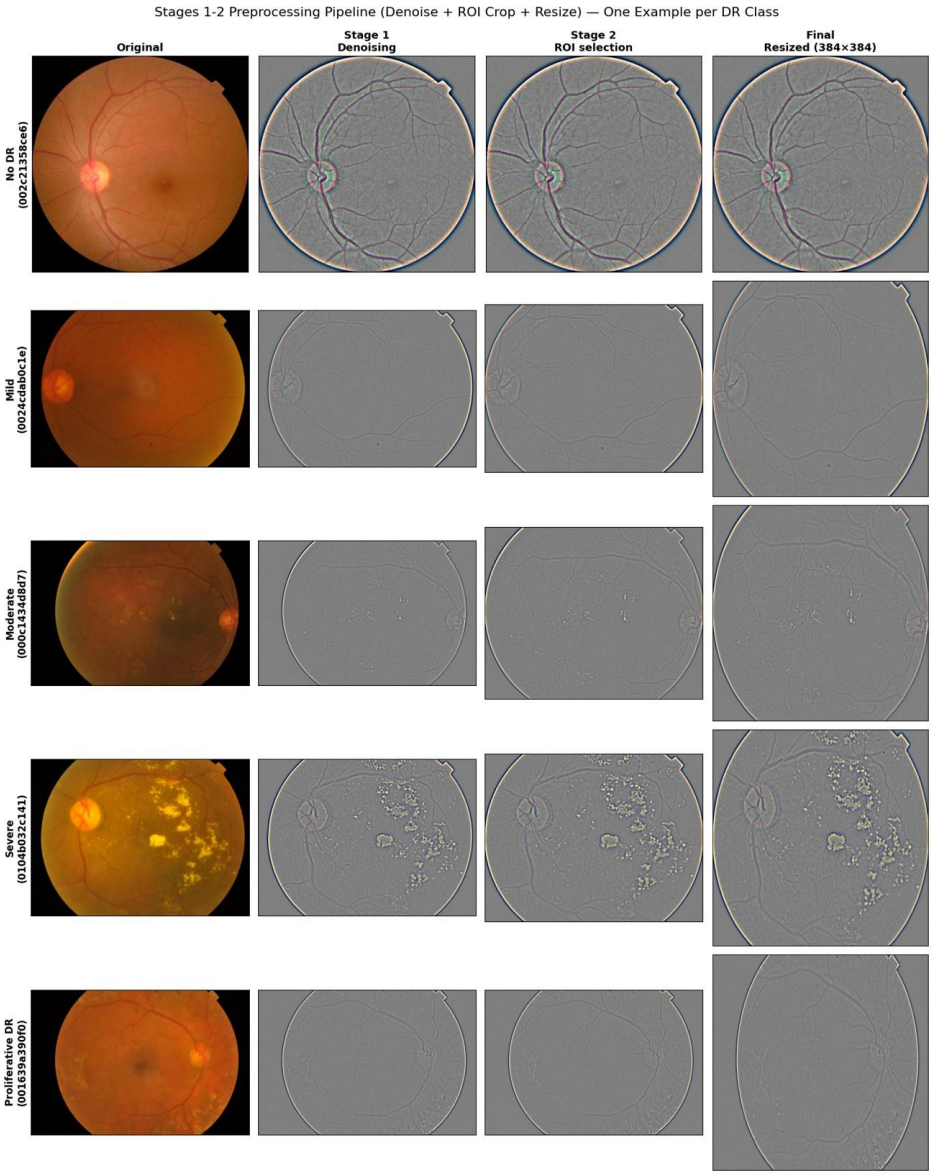


Fig. 4.2. Preprocessing pipeline applied to one APTOS image per DR grade: original image, Stage 1 (denoising), Stage 2 (ROI selection) and the final 384 × 384 image.

4.4.2 Training Augmentation

During training only, the following augmentations are applied to the pre-processed images: random resized crop to 384 × 384 (scale 0.75–1.0, aspect ratio 0.90–1.10), horizontal and vertical flips ($p = 0.5$), shift–scale–rotate (shift 5%, scale ±10%, rotation ±20°, $p = 0.75$), CLAHE ($p = 0.55$), brightness and contrast jitter (±20%, $p = 0.65$), hue–saturation–value jitter ($p = 0.5$), Gaussian noise ($p = 0.2$), Gaussian blur ($p = 0.25$) and coarse dropout of up to eight patches of at most 48 × 48 pixels ($p = 0.35$). Elastic and grid distortions are deliberately excluded because they deform the shape of microaneurysms, which is critical for distinguishing grades. MixUp is disabled because interpolating a grade-0 and a grade-4 image creates a meaningless intermediate grade. The validation images are only normalised.

4.5 Model Training

4.5.1 Model Selection

The teacher must be as accurate as possible, and size is not a constraint, since it runs only on a server. The Swin Transformer [3] was chosen because its shifted-window self-attention models both local lesion texture and global retinal layout at high resolution (384 × 384) at a manageable cost. We use the swin_base_patch4_window12_384 model pre-trained on ImageNet-22K. The student must be small and fast on an ARM processor, so MobileNetV3-Large [15], pre-trained on ImageNet-1k, was chosen. Its depthwise-separable convolutions and Hard-Swish activations are efficient on the Cortex-A72. Both networks are combined with a CBAM attention module [4] so that they can focus on lesion-dense regions.

4.5.2 Model Architecture and Classification Heads

The teacher passes a 3 × 384 × 384 image through the Swin-Base backbone, producing a 12 × 12 × 1024 feature map. The CBAM module then applies channel attention followed by spatial attention. Channel attention passes the global average-pooled and max-pooled descriptors of the feature map F through a shared two-layer MLP (1024 → 64 → 1024), and spatial attention applies a 7 × 7 convolution to the channel-wise average and maximum maps:

$$\mathrm{M_c(F)} = \mathrm{sigmoid}(\ \mathrm{MLP(AvgPool(F))} + \mathrm{MLP(MaxPool(F))}\ ), \quad \mathrm{F'} = \mathrm{M_c(F)} \otimes \mathrm{F} \quad (4.2)$$

$$\mathrm{M_s(F')} = \mathrm{sigmoid}(\ \mathrm{f^{7\times7}}(\ [\mathrm{AvgPool_{ch}(F')}; \mathrm{MaxPool_{ch}(F')}]\ )\ ), \quad \mathrm{F''} = \mathrm{M_s(F')} \otimes \mathrm{F'} \quad (4.3)$$

where F' and F'' are the re-weighted feature maps and $\otimes$ is element-wise multiplication. The attended features pass through global average pooling and a classification head of LayerNorm, dropout (0.35), a fully connected layer (1024 → 512), GELU, dropout (0.35) and a final fully connected layer (512 → 5) that produces the five grade logits. The student has the same overall structure (backbone, CBAM, head) but is much lighter; it is described in Section 4.6.4.

4.5.3 Class Imbalance Handling

Without correction, a model could predict “No DR” for every image and still be about 49% accurate. Imbalance is handled with a weighted random sampler: each training image receives the “balanced” class weight, and 5 × 1,444 = 7,220 images are drawn with replacement per epoch, so that each grade is seen about equally often. Table 4.2 shows how often each

grade's images are repeated. Because the sampler already balances the classes, the loss function is not class-weighted; using both would over-correct.

Table 4.2. Effective draws per epoch after the weighted random sampler (training set).

Grade	Unique images	Draws per epoch	Repetition factor
0 – No DR	1,444	≈ 1,444	1.00×
1 – Mild	296	≈ 1,444	4.88×
2 – Moderate	799	≈ 1,444	1.81×
3 – Severe	154	≈ 1,444	9.38×
4 – Proliferative	236	≈ 1,444	6.12×

4.5.4 Training Strategy

The teacher is trained with focal loss [6], which down-weights easy examples so that training concentrates on hard, often minority-class, images:

$$\mathrm{FL}(p_t) = -(1 - p_t)^{\gamma} \cdot \log(p_t), \quad \gamma = 2 \tag{4.4}$$

where p_t is the predicted probability of the true class. Optimisation uses AdamW [9] with two learning rates: 10^{-5} for the pre-trained backbone and 10^{-4} for the new CBAM module and head, with weight decay 10^{-4}. For the first 5 epochs the backbone is frozen, so that only the randomly initialised CBAM and head are trained; the backbone is then unfrozen for fine-tuning. A cosine-annealing schedule, gradient clipping at 1.0 and automatic mixed precision are used. After each epoch the model is evaluated on the validation set, and the checkpoint with the best QWK is kept. Training stops early if QWK does not improve by at least 0.005 for 10 consecutive epochs. A full checkpoint is written every epoch so that a session interruption can be resumed.

4.5.5 Training Configuration

Table 4.3 lists the configuration of the teacher run and of the student distillation run.

Table 4.3. Training configuration of the teacher and the student.

Setting	Teacher (Phase A)	Student (Phase B/C)
Input size	384 × 384	384 × 384 batch, resized to 224 × 224 inside the model
Maximum epochs	100	50
Early-stopping patience	10 epochs on validation QWK (Δ ≥ 0.005)	10 epochs on validation QWK (Δ ≥ 0.005)
Optimiser	AdamW, weight decay 1e-4	AdamW, weight decay 1e-4
Learning rate	1e-5 backbone; 1e-4 CBAM + head	1e-3 (all parameters)
LR schedule	Cosine annealing	Cosine annealing to 1e-6
Backbone warm-up	5 epochs frozen	None
Batch size	8 (validation 4)	32 (validation 16)

Setting	Teacher (Phase A)	Student (Phase B/C)
Loss	Focal loss, $\gamma = 2$	$0.7 \cdot T^2 \cdot KL + 0.3 \cdot CE + 0.1 \cdot MSE$, $T = 4$
Class balancing	Weighted random sampler	Weighted random sampler
Precision / clipping	Mixed precision; gradient clip 1.0	Mixed precision; gradient clip 1.0
Trainable parameters	87.5 M	3.05 M (+ projector)

4.6 Proposed Knowledge-Distillation Framework

4.6.1 Overview

The proposed framework (Fig. 4.3) has four phases. In Phase A a large teacher is trained for accuracy on a GPU server. In Phase B the teacher's knowledge is transferred to a small student through a composite distillation loss. Phase C defines the edge student architecture. In Phase D the student is exported to ONNX and quantised to INT8 for the Raspberry Pi. Fig. 4.4 shows the detailed block architecture of the teacher and the student.

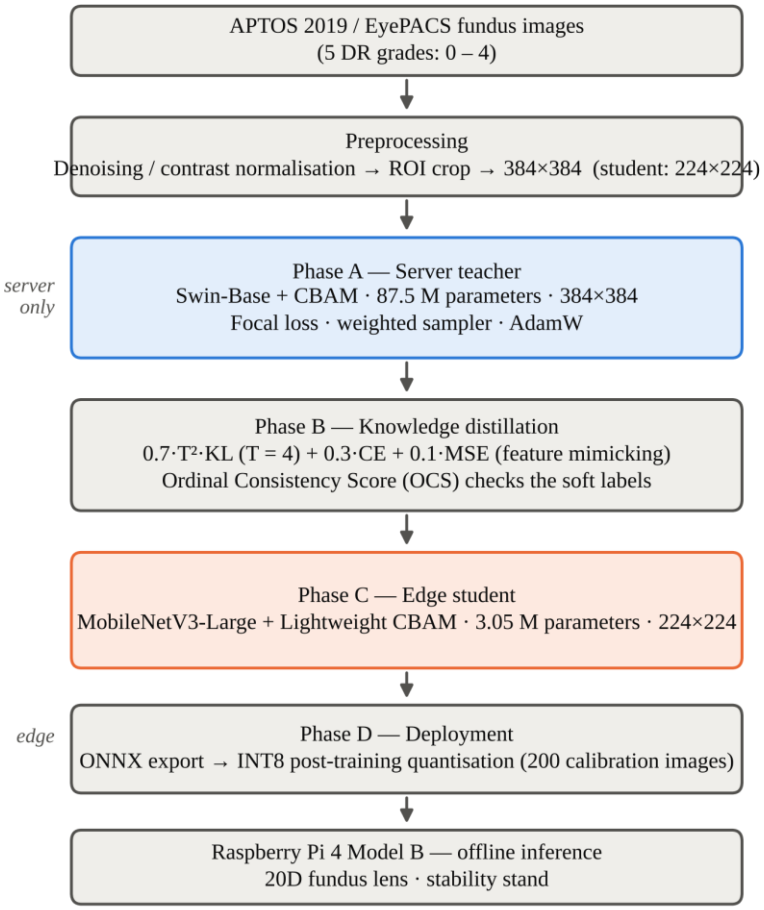


Fig. 4.3. Overall OpthaEdge-DED pipeline: Phase A (teacher), Phase B (knowledge distillation), Phase C (edge student) and Phase D (ONNX / INT8 deployment).

4.6.2 Phase A: Server-Side Teacher Training

The teacher is the Swin-Base + CBAM network of Section 4.5, with 87.5 million parameters. It is trained as described in Section 4.5.4 and then frozen. It provides the "soft labels" (class probability vectors) and the pooled backbone features used in Phase B.

4.6.3 Phase B: Knowledge Distillation

Knowledge distillation [10] trains the student on the teacher's softened probability distribution as well as on the true labels. The soft distribution contains "dark knowledge": it shows which wrong grades the teacher finds similar to the right one. For ordered DR grades, a good teacher should give more probability to neighbouring grades than to distant ones. The student is trained with the composite loss

$$\mathrm{L} = 0.7 \cdot \mathrm{T}^2 \cdot \mathrm{KL}(\, \mathrm{S}(z_T/\mathrm{T}) \,\|\, \mathrm{S}(z_S/\mathrm{T})\,) + 0.3 \cdot \mathrm{CE}(z_S, \mathrm{y}) + 0.1 \cdot \mathrm{MSE}(\, \mathrm{W_p} f_T, f_S\,) \qquad (4.5)$$

where z_T and z_S are the teacher and student logits, $S(\cdot)$ is the softmax, y is the true grade, and the temperature $T = 4$ softens the distributions. The factor T^2 restores the gradient magnitude lost by softening. The first term (weight 0.7) transfers the teacher's soft labels, the second (weight 0.3) keeps the student anchored to the ground truth, and the third (weight 0.1) is a feature-mimicking loss between the global-average-pooled backbone features of the teacher (f_T, 1024-d) and the student (f_S, 960-d). Because the dimensions differ, a learnable linear projector W_p (1024 → 960) is trained together with the student and discarded afterwards. The teacher is frozen throughout, and the same augmented batch is given to both networks, each at its own resolution.

4.6.4 Phase C: Edge Student Architecture

The student is built for ARM execution (Table 4.4, Fig. 4.4). Its backbone is MobileNetV3-Large, which maps a 224 × 224 image to a 960 × 7 × 7 feature map. A **lightweight CBAM** follows. It keeps the two-step attention of Eqs. (4.2)–(4.3) but reduces the cost in two ways: the channel MLP uses a 24× bottleneck (960 → 40 → 960), and the spatial branch replaces the dense 7 × 7 convolution with a depthwise 7 × 7 convolution followed by a 1 × 1 pointwise convolution. Global average pooling and a single fully connected layer (960 → 5) produce the logits. The student has 3.05 million parameters (backbone 2.97 M, lightweight CBAM 76,900, classifier 4,805), which is 28.7 times fewer than the teacher. The unused 960

→ 1280 convolution that the timm library keeps for its own classifier is removed, so the parameter count reflects only what the student runs.

Table 4.4. Teacher and student architectures.

Property	Teacher (Phase A)	Student (Phase C)
Backbone	Swin-Base (ImageNet-22K)	MobileNetV3-Large (ImageNet-1k)
Input resolution	384 × 384	224 × 224
Backbone output	12 × 12 × 1024	7 × 7 × 960
Attention module	CBAM: 1024 → 64 → 1024; dense 7 × 7 spatial conv	Lightweight CBAM: 960 → 40 → 960; depthwise 7 × 7 + 1 × 1 spatial conv
Attention parameters	131,170	76,900
Classification head	GAP → LayerNorm → FC 512 → GELU → FC 5 (529,413 parameters)	GAP → FC 5 (4,805 parameters)
Total parameters	87.5 M	3.05 M
Role	Server-side training only	Edge deployment

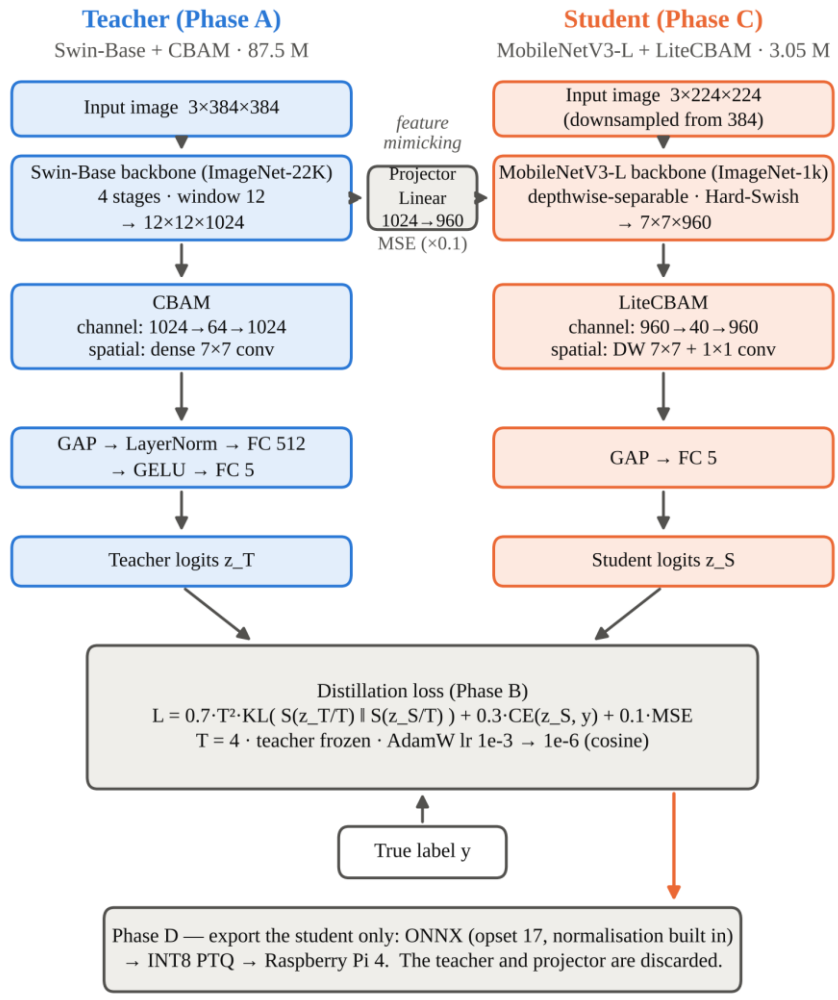


Fig. 4.4. Detailed block architecture of the teacher (Phase A) and student (Phase C), and the distillation losses connecting them (Phase B).

4.6.5 Phase D: Quantisation and Deployment Strategy

After distillation, only the student is exported to ONNX (opset 17) with a dynamic batch dimension. The ImageNet normalisation is folded into the exported graph, so the deployed model takes RGB float32 images in the range [0, 1] of shape $N \times 3 \times 224 \times 224$ and returns five logits. On the device the pipeline is therefore: preprocess the captured image (Section 4.4.1), resize to 224 × 224, divide by 255 and run the network. The exported model is checked against PyTorch on real validation images before quantisation.

The model is then quantised with ONNX Runtime [18] static post-training quantisation in the QDQ format, using per-channel INT8 weights and UINT8 activations. Activation ranges are calibrated on 200 training images, 40 per DR grade, so that the rare severe grades influence the ranges as much as grade 0. INT8 storage is about four times smaller than FP32, and ONNX Runtime can use NEON integer instructions on the Cortex-A72. The quantised model runs fully offline on the Raspberry Pi 4.

4.7 Ordinal Consistency Score (OCS)

Accuracy and QWK evaluate only the predicted grade, but distillation transfers the whole probability vector. To measure whether ordinal structure survives distillation, we define the Ordinal Consistency Score of a probability vector $\mathbf{p}$ over K ordered grades. Let k = argmax $\mathbf{p}$ be the predicted grade. Moving away from k in either direction, the probability should never increase:

$$\mathrm{OCS}(\mathrm{p}) = 1/(\mathrm{K}-1) \cdot [\, \Sigma_{i<k}\, 1(\, p_i \le p_{i+1}\,) + \Sigma_{i \ge k}\, 1(\, p_i \ge p_{i+1}\,)\,], \quad i = 0, \ldots, K-2 \qquad (4.6)$$

OCS lies in [0, 1], and 1 means that the distribution is perfectly unimodal around the predicted grade. For example, $\mathbf{p}$ = (0.05, 0.15, 0.50, 0.25, 0.05) has OCS = 1, whereas $\mathbf{p}$ = (0.30, 0.05, 0.40, 0.20, 0.05) has OCS = 0.75, because grade 0 is more probable than grade 1 although grade 1 is nearer to the peak. Only the order of the probabilities matters, so OCS does not depend on the softmax temperature. The mean OCS over the validation set is reported for the teacher and the student. The metric was checked against hand-computed cases in a unit test.

4.8 Evaluation Metrics

Models are compared with accuracy, macro-averaged precision, recall and F1-score, and the quadratic weighted kappa (QWK). QWK [5] is the primary metric for ordinal grading, since it penalises a wrong prediction by the squared distance between the true and predicted grades:

$$\kappa = 1 - (\, \Sigma_{i,j}\, w_{ij}\, O_{ij}\,) \,/\, (\, \Sigma_{i,j}\, w_{ij}\, E_{ij}\,), \quad w_{ij} = (i-j)^2 \,/\, (K-1)^2 \qquad (4.7)$$

where O is the observed confusion matrix and E the matrix expected by chance. Macro averaging gives each grade equal weight, which matters because the rare grades are clinically important. The confusion matrix and per-grade results are also reported. Deployment is characterised by model size, accuracy after quantisation, and mean single-image latency on the Raspberry Pi 4.

4.9 Implementation Environment

Model training was run in Google Colab on a Tesla T4 GPU (15 GB) with Python 3.13, using PyTorch, the timm library for the pre-trained backbones, Albumentations for augmentation, scikit-learn for the metrics, and ONNX and ONNX Runtime for export and quantisation. Data and checkpoints were stored on Google Drive, and training resumes automatically from the latest checkpoint. All random seeds were fixed to 42. The deployed model runs on the Raspberry Pi 4 Model B with ONNX Runtime.

4.10 OpthaEdge Web Platform

4.10.1 Purpose and Architecture

The trained models are intended to be used by clinicians and screening staff, not as scripts. For this reason the project includes a web application, the OpthaEdge clinical workspace, which wraps the model in a complete screening workflow: patient registration, image capture or upload, image-quality checking, AI-assisted grading, clinician review, record keeping and report generation. The application is a browser-based single-page interface built with Next.js and React, with a dark clinical theme for low-light examination rooms, and it runs on the same local network as the Raspberry Pi imager so that fundus images and patient data stay inside the clinic. The web server calls the exported ONNX models through ONNX Runtime, so the application uses exactly the deployed artefact evaluated in Chapter 5.

The platform is organised in three layers. The **presentation layer** is the browser interface described below. The **application layer** holds the authentication and role-based access control, the screening workflow, the image-quality gate, the audit log and the report generator. The **inference and data layer** holds the preprocessing code of Section 4.4, the ONNX models with a versioned model registry, and the encrypted storage of images and records.

Table 4.5. Modules of the OpthaEdge clinical workspace.

Module	Function
Overview	Dashboard with the number of patients, screenings performed, ungradable images and active research models, with quick actions to add a patient or start a screening.
Patients	Registration of patients with consent, demographics and diabetes history; search and per-patient history.
New Screening	Choice of analysis mode, image upload or capture from the Raspberry Pi imager, image-quality check, inference and result display.
Examinations	Immutable, eye-specific (left/right) records holding the original image, quality assessment, model version, prediction, clinician review and audit history.
Disease Library	Patient-friendly education on DR and other eye diseases, with versioned sources and a named clinical reviewer for every article.
Reports	Print-friendly and PDF screening reports generated only from stored, authorised records, with the AI result clearly separated from the clinician's review.
Administration	Roles and permissions, retention and consent policies, model approvals, source reviews and an auditable configuration history.
Model Information	Registry of the deployed models: artefact, input and output shape, preprocessing version, label order, known limitations and validation status.

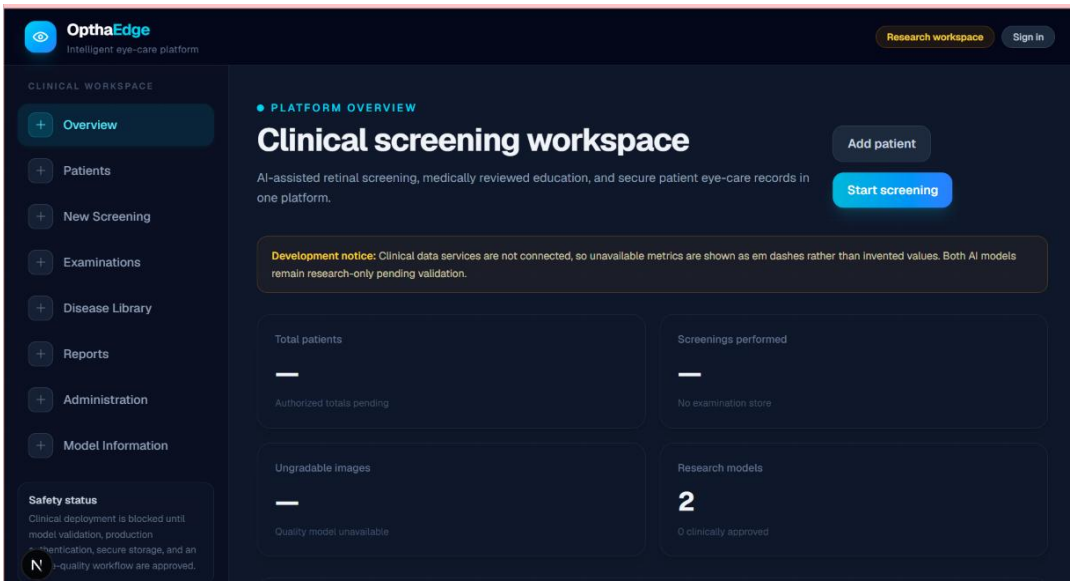


Fig. 4.5. Overview page of the OpthaEdge clinical workspace, with the module navigation on the left and the safety status panel.

4.10.2 Screening Workflow

A screening follows the steps below. The user chooses a patient and an analysis mode on the New Screening page (Fig. 4.6): **DR / non-DR detection**, a binary screening model, or **DR severity grading**, the five-grade classifier of this report. The fundus image is then checked for quality. The image is rejected and a re-capture is requested if it is blurred, poorly illuminated or off-centre, which avoids a grade being assigned to an ungradable image. A good image goes through the Stage 1 and ROI preprocessing of Section 4.4 and is passed to the ONNX student model, which returns five logits. The interface converts these to class probabilities and shows

the predicted grade with its probability distribution, so that the user can see when the model is uncertain between two neighbouring grades; this is where the ordinal structure measured by the OCS becomes visible to the user. The clinician then confirms or overrides the grade, and the final decision is stored together with the model and preprocessing versions in an immutable examination record.

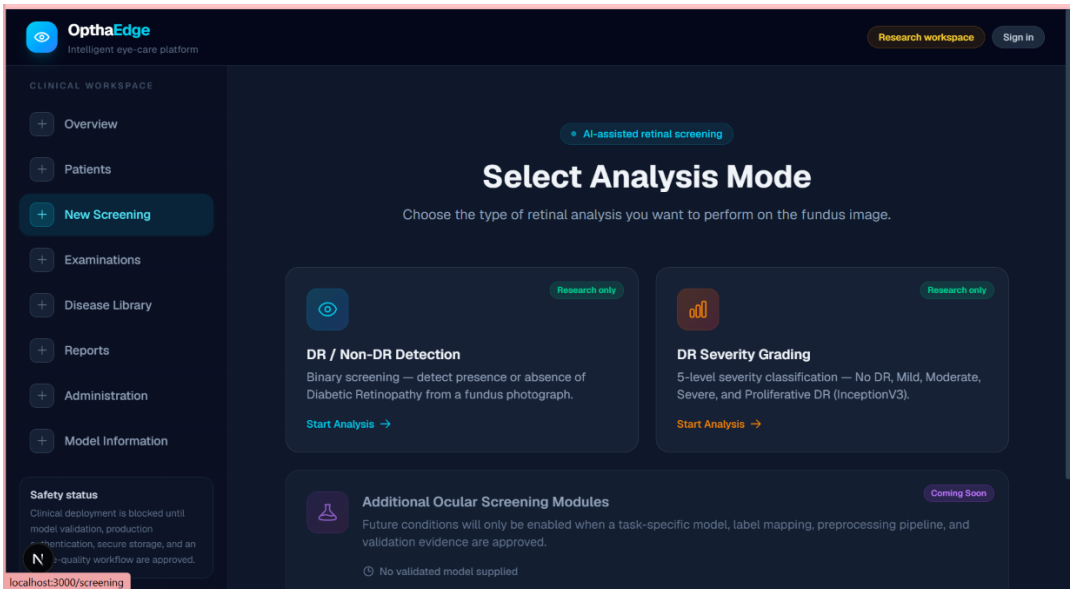


Fig. 4.6. New Screening page showing the two analysis modes (DR / non-DR detection and DR severity grading); further ocular modules are listed as coming soon.

4.10.3 Safety, Governance and Privacy

Because the output of the platform concerns a medical condition, safety is built into the design and not added afterwards. Every page carries a permanent safety panel stating that the models are for research and screening support only, and are not approved for diagnosis. Users must sign in (Fig. 4.7) through an OpenID Connect identity provider, have an active organisation membership and hold the permissions needed for each action, and these permissions are enforced on the server and not only in the interface. Patient images are stored encrypted, every access and configuration change creates an audit event, and retention and consent policies are set by the administrator. Where a value is not available, the interface shows an empty marker and never an invented number.

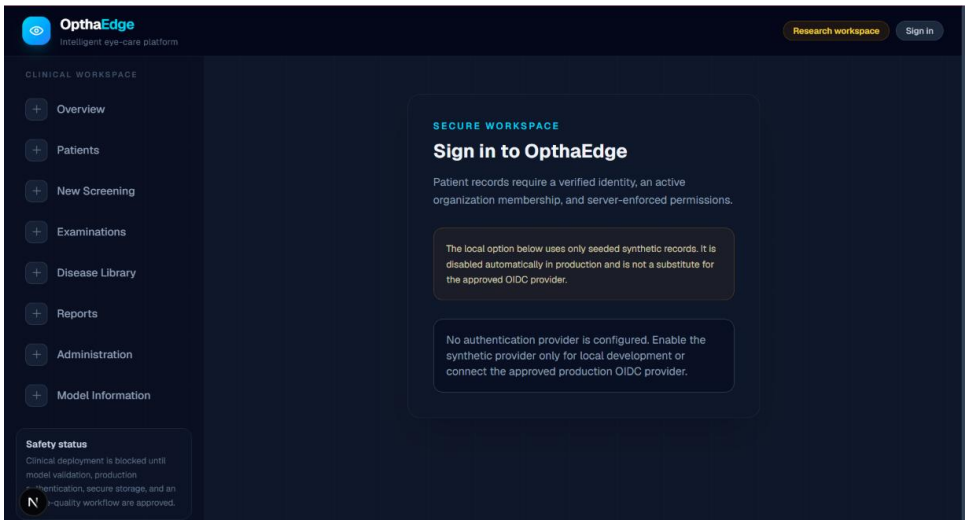


Fig. 4.7. Sign-in page: access to patient records needs a verified identity, an active organisation membership and server-enforced permissions.

The Model Information page (Fig. 4.8) is the governance record of the platform. For each model it lists the ONNX artefact, the input and output shapes, the preprocessing version and the label assumption, and it states the validation status and the known limitations. A model is marked as blocked until its training configuration, label order, preprocessing provenance, calibration, threshold selection, validation datasets, performance metrics, confidence intervals and subgroup results are on record, and it cannot be used for clinical screening until it is approved by the administrator. The distilled student of this report is registered here with its INT8 and FP32 artefacts, its 224 × 224 input contract (RGB, range [0, 1], normalisation inside the graph) and the validation results of Chapter 5.

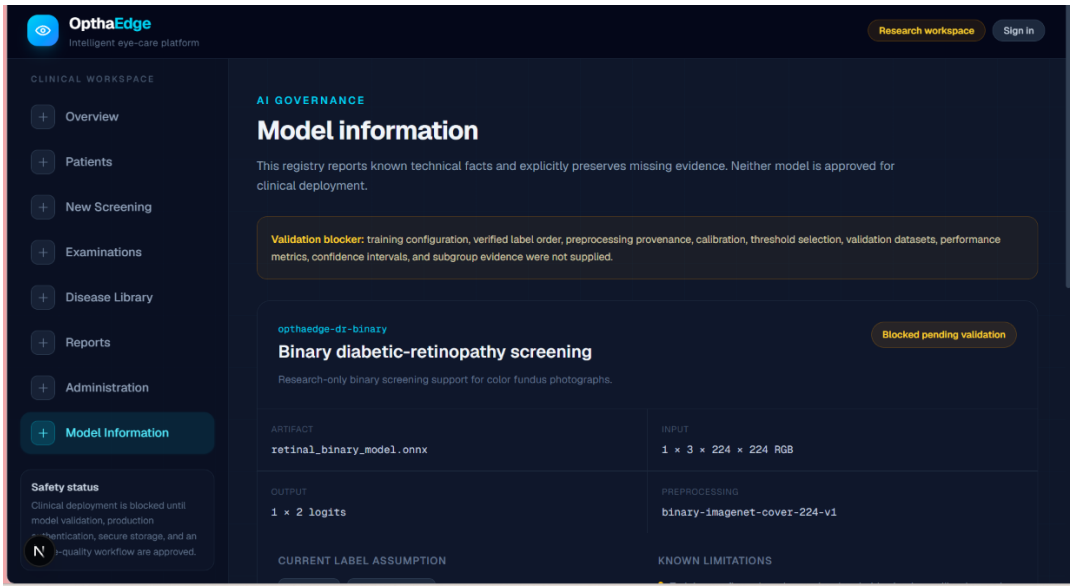


Fig. 4.8. Model Information page: artefact, input, output and preprocessing of the registered model, and its validation status.

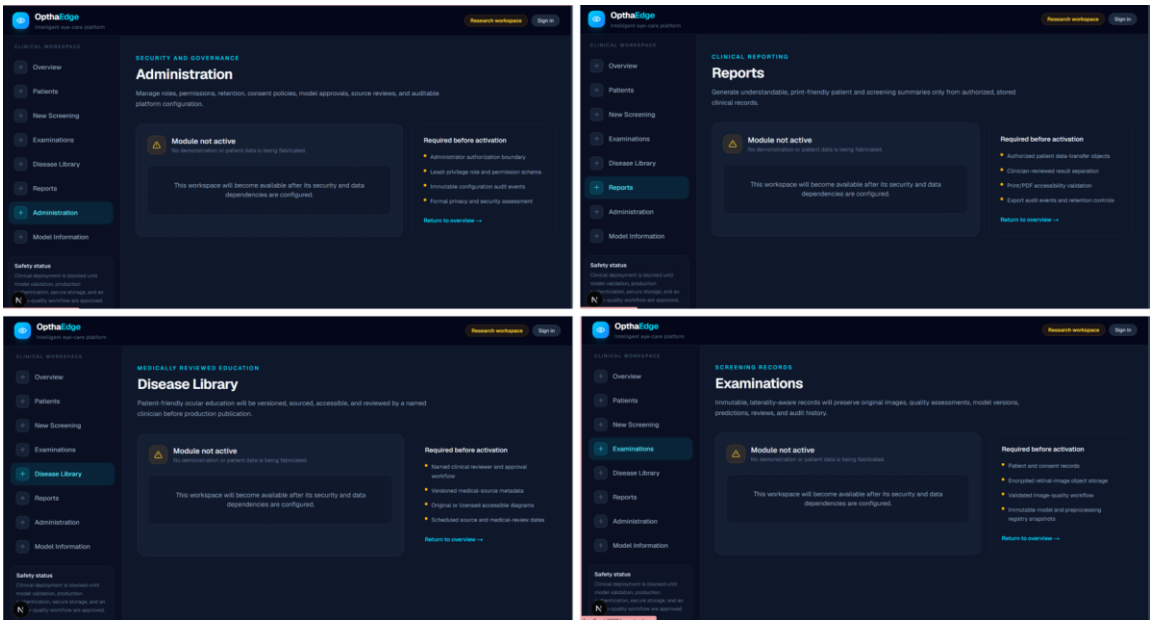


Fig. 4.9. Other modules: (a) Administration, (b) Reports, (c) Disease Library and (d) Examinations.

The screenshots in this section were taken from the development build of the platform running on a local server. In this build the backend data services and the identity provider are not connected, and the interface therefore shows the empty markers and “module not active” messages described above in place of patient data. No demonstration or patient data were fabricated for it. The functions described in this section are those intended for the complete platform.

Chapter 5

Results and Discussion

5.1 Overview of Experimental Analysis

This chapter reports the experiments carried out so far. All results are on the held-out 20% validation split of APTOS 2019 (733 images; 361, 74, 200, 39 and 59 images for grades 0 to 4). The teacher (Phase A), the distillation of the student (Phases B and C) and the ONNX/INT8 export (Phase D) have all been run, and the numbers below are taken directly from the notebook output. Only the measurements on the Raspberry Pi itself are still missing, and these are shown in **yellow-highlighted** cells, to be filled in from the device.

The validation split was used both to choose the best checkpoint (by QWK) and to report the results; no separate test split was held out. The reported numbers are therefore slightly optimistic, and this is discussed in Section 5.7.

5.2 Teacher Performance (Phase A)

5.2.1 Overall Results

The Swin-Base + CBAM teacher reached an accuracy of 82.13%, a QWK of 0.9043 and a macro F1-score of 0.6820 on the validation set (Table 5.1). The high QWK relative to the accuracy shows that most errors are small ones, between neighbouring grades. Training stopped early at epoch 19, because QWK had not improved for 10 epochs, and the best checkpoint (epoch 9) was kept.

Table 5.1. Teacher performance on the validation set (best checkpoint).

Metric	Value	Note
Accuracy	0.8213	602 of 733 correct
Quadratic weighted kappa (QWK)	0.9043	Primary metric
Macro precision	0.6871	
Macro recall	0.6841	
Macro F1-score	0.6820	
Validation loss (focal)	0.3077	
Parameters	87.5 M	

5.2.2 Training Behaviour

Fig. 5.1 shows the training history. The training loss falls steadily from 0.70 to 0.12 over 19 epochs, but the validation loss reaches its minimum at epoch 7 (0.245) and then rises to 0.61, while the validation accuracy and QWK stay roughly flat at 0.79–0.83 and 0.88–0.90. This is a clear sign of over-fitting after about epoch 9: the model keeps fitting the training images without generalising better. Early stopping and checkpoint selection on QWK prevented the over-fitted weights from being used. With 2,929 training images and 87.5 million parameters, this behaviour is expected.

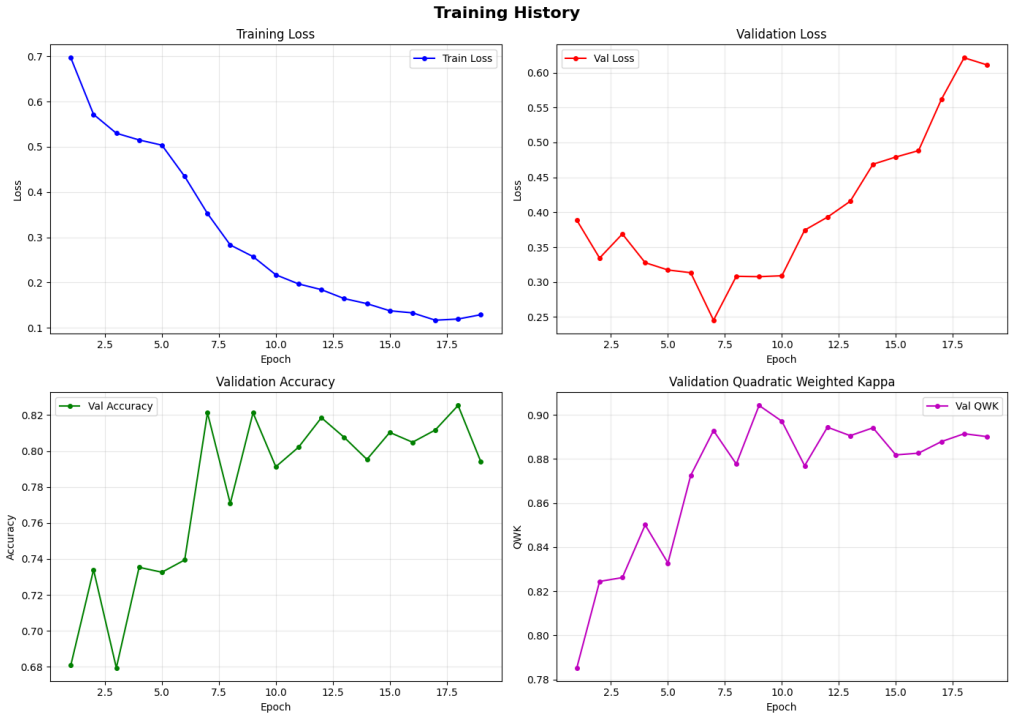


Fig. 5.1. Teacher training history: training loss, validation loss, validation accuracy and validation QWK per epoch.

5.2.3 Per-Grade Results and Error Analysis

Table 5.2 and Figs. 5.2 and 5.3 give the per-grade results. The two extreme classes behave very differently. Grade 0 is recognised almost perfectly (precision 0.986, recall 0.945), but the rarest grade, severe DR (grade 3, only 154 training images), has the lowest F1-score of 0.42. Mild DR is also hard (F1 0.62), as its lesions are few and small.

Table 5.2. Per-grade results of the teacher on the validation set.

Grade	Precision	Recall	F1-score	Support
0 – No DR	0.9855	0.9446	0.9646	361
1 – Mild	0.5833	0.6622	0.6203	74
2 – Moderate	0.7642	0.8100	0.7864	200
3 – Severe	0.3913	0.4615	0.4235	39

Grade	Precision	Recall	F1-score	Support
4 – Proliferative	0.7111	0.5424	0.6154	59
Macro average	0.6871	0.6841	0.6820	733
Weighted average	0.8308	0.8213	0.8243	733

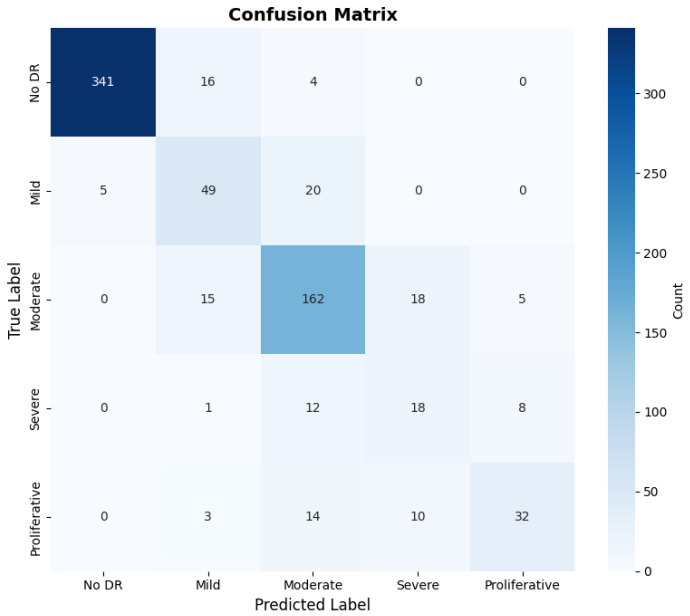


Fig. 5.2. Confusion matrix of the teacher on the validation set (rows: true grade; columns: predicted grade).

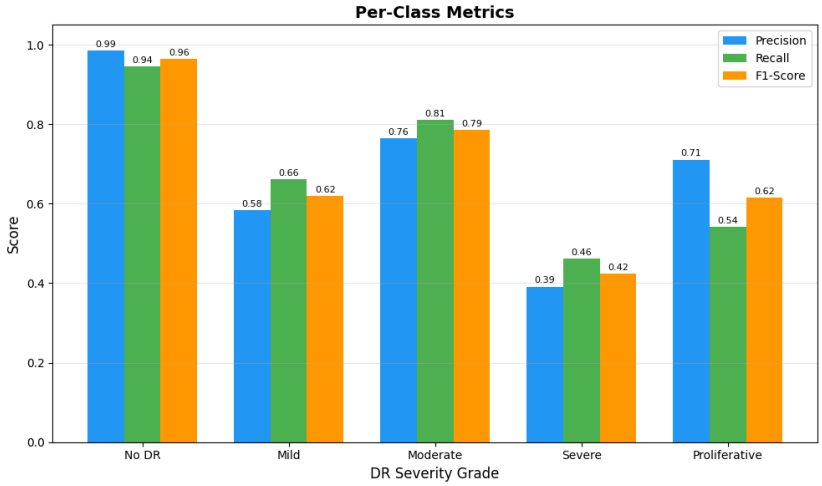


Fig. 5.3. Per-grade precision, recall and F1-score of the teacher.

The confusion matrix (Fig. 5.2) shows that the model's mistakes are mostly small. Of the 131 wrong predictions, 104 (79.4%) are off by one grade, 24 (18.3%) by two grades and only 3 (2.3%) by three grades; no grade-0 image was ever predicted as proliferative DR, or the reverse. The main confusions are mild ↔ moderate (35 images), moderate ↔ severe (30) and proliferative → moderate (14). Reading the same matrix as a screening decision gives a sensitivity of 98.7% and a specificity of 94.5% for detecting any DR (grade ≥ 1), and a

sensitivity of 93.6% (279 of 298) and a specificity of 94.5% (411 of 435) for referable DR (grade ≥ 2). These screening figures are derived from the confusion matrix and are not a separate experiment.

5.3 Student Performance after Distillation (Phases B and C)

5.3.1 Distillation Progress

The MobileNetV3-Large + lightweight-CBAM student was distilled from the frozen teacher using the composite loss of Eq. (4.5). Table 5.3 and Fig. 5.4 show the progress. All loss terms fall steadily: the soft-label KL term from 0.97 to 0.19, the hard-label cross-entropy from 0.96 to 0.40 and the feature-mimicking MSE from 0.139 to 0.021. The student learns quickly. After one epoch its validation QWK was already 0.870, and the saved checkpoint (QWK 0.9033, accuracy 0.8145) was reached at epoch 6. QWK reached 0.9047 at epoch 13, but this is less than 0.005 above the saved value and was not counted as an improvement. Training stopped early at epoch 19, when the patience of 10 epochs had run out.

Table 5.3. Selected epochs of the student distillation run (validation set).

Epoch	Soft KL	Hard CE	Feat. MSE	Val. acc.	Val. QWK	Val. F1	Note
1	0.967	0.964	0.139	0.7231	0.8695	0.5727	First epoch
2	0.538	0.786	0.073	0.7858	0.8784	0.6256	Saved
6	0.315	0.550	0.041	0.8145	0.9033	0.6628	**Best, saved**
10	0.260	0.482	0.032	0.7995	0.9003	0.6556	
13	0.237	0.446	0.028	0.8131	0.9047	0.6639	$\Delta < 0.005$
19	0.192	0.396	0.021	0.8158	0.8963	0.6608	Early stop

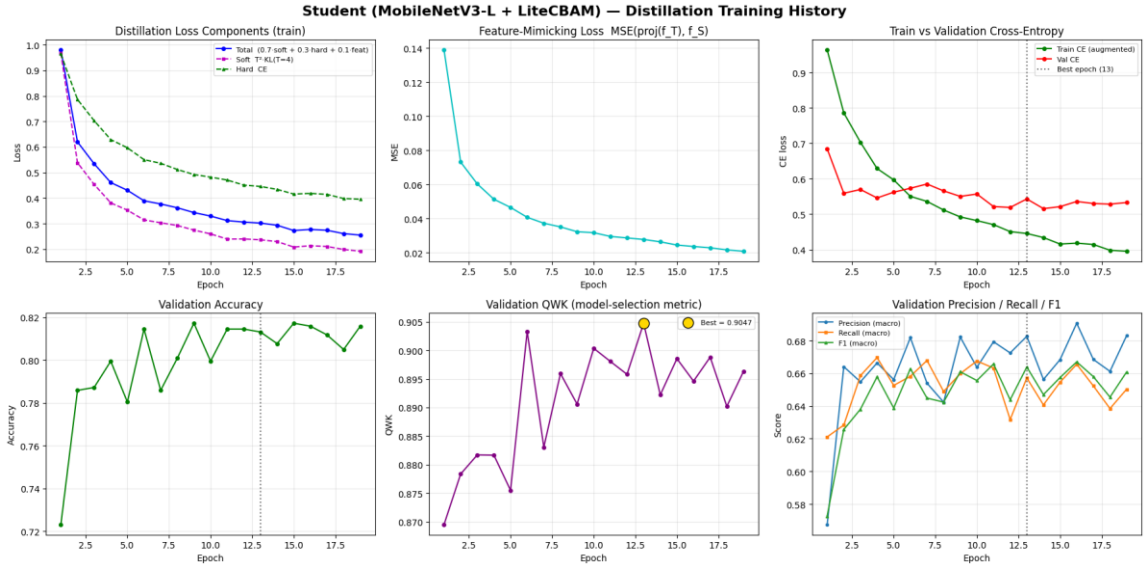


Fig. 5.4. Student distillation history: loss components, feature-mimicking loss, train and validation cross-entropy, validation accuracy, QWK and macro precision/recall/F1.

5.3.2 Teacher versus Student

Table 5.4 and Fig. 5.5 compare the student with the teacher. With 28.7 times fewer parameters, the student keeps 99.2% of the teacher's accuracy (0.8145 against 0.8213), 99.9% of its QWK (0.9033 against 0.9043) and 97.2% of its macro F1-score (0.6628 against 0.6820). The student's epoch-to-epoch QWK varies by about ±0.006 (standard deviation over epochs 6–16), which is larger than the 0.001 gap to the teacher, so the two models should be regarded as equal in QWK. The student is slightly weaker in recall (0.658 against 0.684), which is dominated by the rare grades.

Table 5.4. Comparison of the teacher and the distilled student on the validation set.

Metric	Teacher	Student (KD)	Student / teacher
Parameters	87.5 M	3.05 M	1 / 28.7
Input resolution	384 × 384	224 × 224	
Accuracy	0.8213	0.8145	99.2%
QWK	0.9043	0.9033	99.9%
Macro precision	0.6871	0.6818	99.2%
Macro recall	0.6841	0.6582	96.2%
Macro F1-score	0.6820	0.6628	97.2%
Ordinal Consistency Score (OCS)	0.7954	0.7950	99.9%

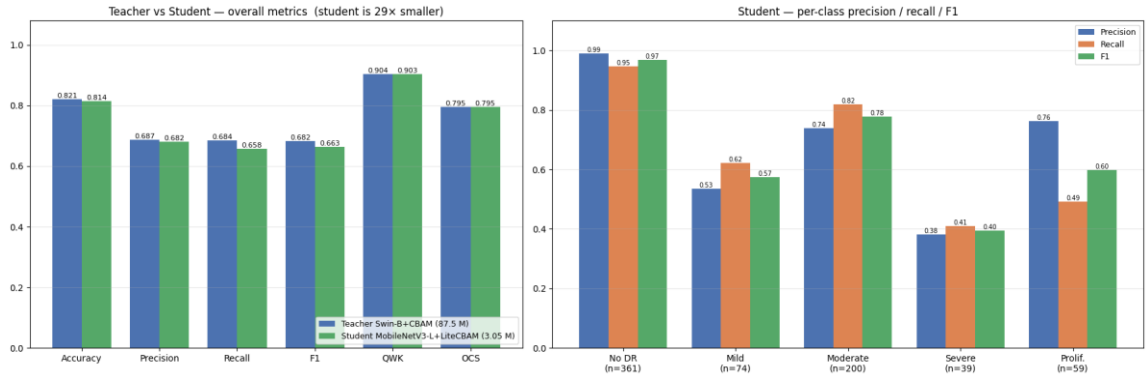


Fig. 5.5. (a) Overall metrics of the teacher and the distilled student; (b) per-grade precision, recall and F1-score of the student.

5.3.3 Per-Grade Results of the Student

Table 5.5 and Fig. 5.6 give the per-grade results of the student. The pattern matches the teacher: grade 0 is recognised almost perfectly (F1 0.969), moderate DR is good (0.777), and the rare grades are weakest, with severe DR at F1 0.395 and proliferative DR at 0.598. Proliferative DR has high precision (0.763) but low recall (0.492); half of these images are predicted as moderate (16) or severe (11) DR. Of the student's 136 errors, 112 (82.4%) are off by one grade, 21 (15.4%) by two and 3 (2.2%) by three. Read as a screening test, the student detects 99.2% of DR cases (369 of 372) with a specificity of 94.7% (342 of 361), and detects

91.9% of referable DR cases (274 of 298) with a specificity of 93.6% (407 of 435). The macro one-vs-rest AUC is 0.942 (Fig. 5.8).

Table 5.5. Per-grade results of the distilled student on the validation set.

Grade	Precision	Recall	F1-score	Support
0 – No DR	0.9913	0.9474	0.9688	361
1 – Mild	0.5349	0.6216	0.5750	74
2 – Moderate	0.7387	0.8200	0.7773	200
3 – Severe	0.3810	0.4103	0.3951	39
4 – Proliferative	0.7632	0.4915	0.5979	59
Macro average	0.6818	0.6582	0.6628	733
Weighted average	0.8255	0.8145	0.8164	733

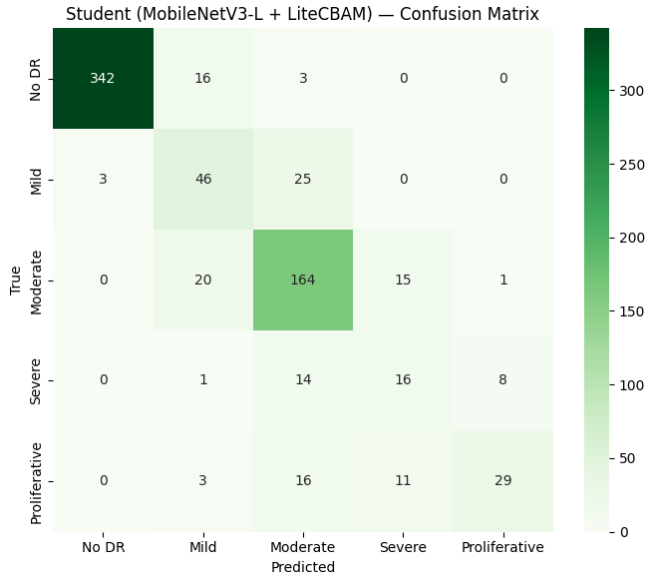


Fig. 5.6. Confusion matrix of the distilled student on the validation set (rows: true grade; columns: predicted grade).

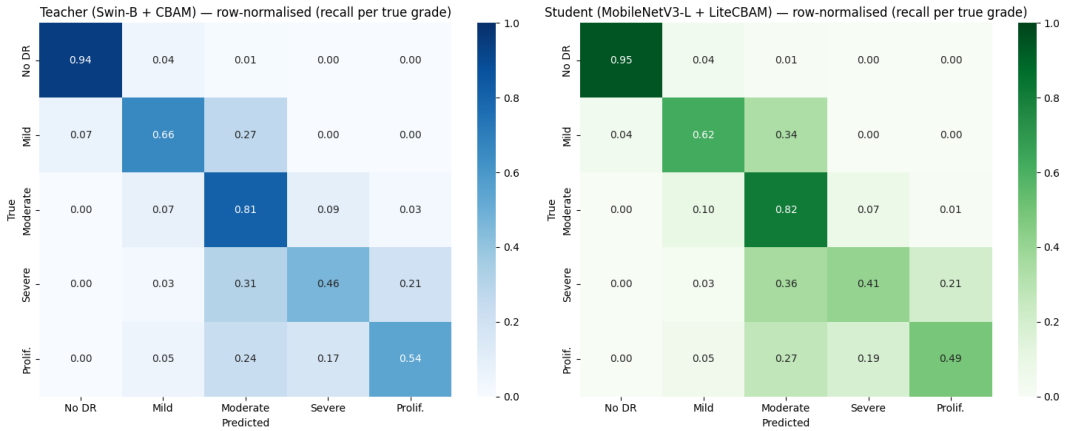


Fig. 5.7. Row-normalised confusion matrices (recall per true grade): (a) teacher, (b) student.

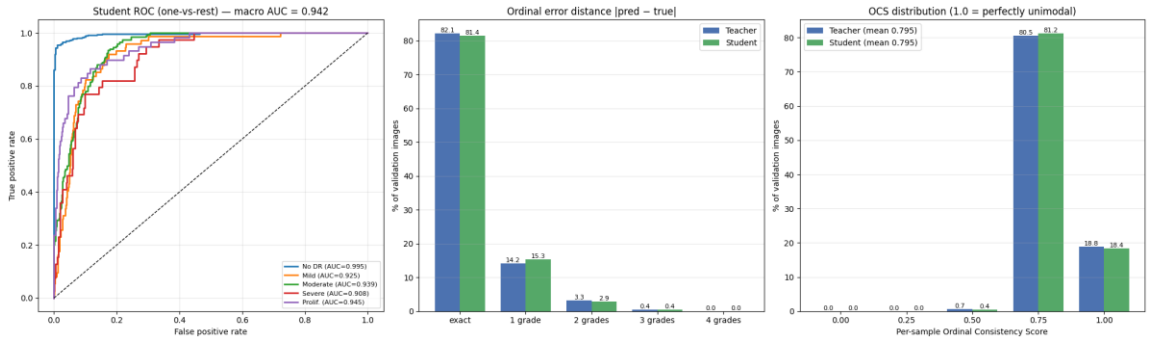


Fig. 5.8. (a) One-vs-rest ROC curves of the student (macro AUC 0.942); (b) distribution of ordinal error distance; (c) distribution of the per-sample OCS.

5.3.4 Ordinal Consistency

The OCS of the teacher is 0.7954 and that of the student is 0.7950, so the student has inherited the ordinal structure of the teacher almost exactly. The per-sample OCS is concentrated in two bins (Fig. 5.8c): about 81% of images score in the 0.75 bin and about 18–19% score 1.0 (perfectly unimodal), and very few fall below 0.5, for both models. The averaged soft labels (Fig. 5.9) show the same effect: for every true grade, probability is highest at the true grade and in the neighbouring grades, and the student's heat map is nearly identical to the teacher's. The distribution of the ordinal error distance is also almost the same (exact 82.1% against 81.4%; one grade 14.2% against 15.3%). The OCS is not a clinical measure, but its closeness for the two models shows that the distillation transferred the teacher's ordered view of the grades and not only its top-1 decisions.

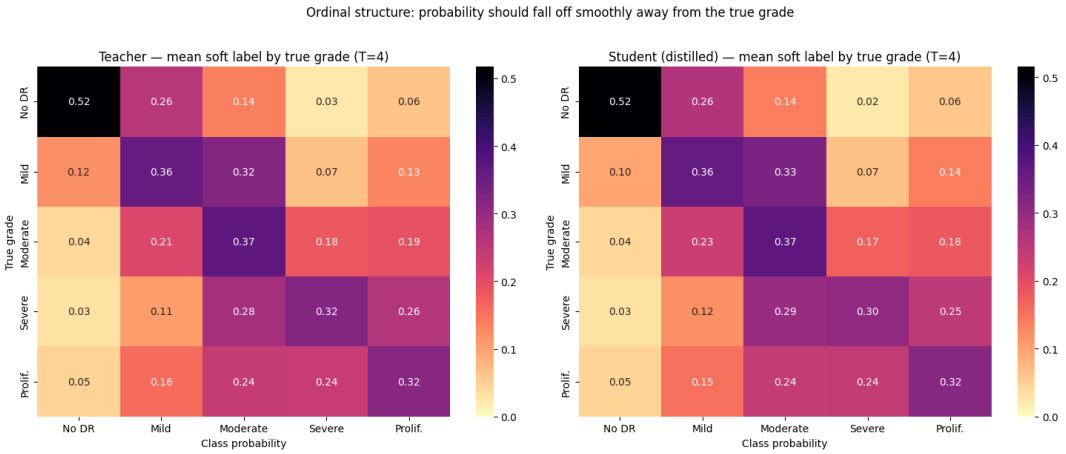


Fig. 5.9. Mean soft label (T = 4) by true grade: (a) teacher, (b) distilled student. Probability should fall off smoothly away from the true grade.

5.4 Deployment Results (Phase D)

5.4.1 ONNX Export and INT8 Quantisation

The exported teacher ONNX file is 373.34 MB, which is far too large for a Raspberry Pi. The student exported to ONNX (opset 17) is 12.21 MB in FP32 and 3.55 MB after static

INT8 quantisation, which is 3.4 times smaller than the FP32 file and 105 times smaller than the teacher. The FP32 graph reproduced the PyTorch logits to within 3.3×10^{-6} and gave the same predictions on all 16 real validation images tested. Table 5.6 and Figs. 5.10 and 5.11 compare the two student models.

Table 5.6. Size, accuracy and latency of the exported models.

Metric	Teacher ONNX	Student FP32 ONNX	Student INT8 ONNX
File size (MB)	373.34	12.21	3.55
Input	384 × 384 (normalised)	224 × 224, [0, 1]	224 × 224, [0, 1]
Validation accuracy	0.8213	0.8145	0.7844
Validation QWK	0.9043	0.9033	0.8900
Macro F1-score	0.6820	0.6628	0.6355
OCS	0.7954	0.7950	0.7981
FP32–INT8 prediction agreement	–	–	89.5%
Latency on x86 CPU, 1 thread (ms/image)	Not deployable	6.8	10.3
Latency on Raspberry Pi 4 (ms/image)	Not deployable	[measure]	[measure]
Peak memory on Raspberry Pi 4 (MB)	Not deployable	[measure]	[measure]

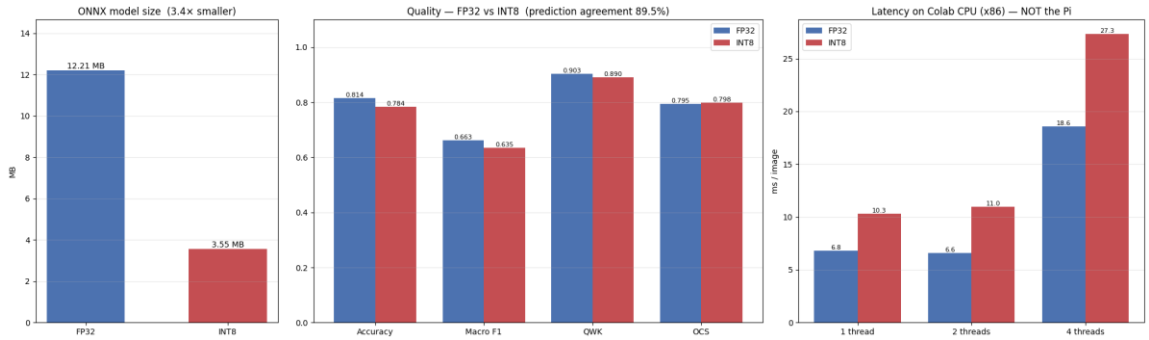


Fig. 5.10. (a) ONNX model size, (b) quality of the FP32 and INT8 student models, (c) latency on the Colab x86 CPU for 1, 2 and 4 threads (not the Raspberry Pi).

Quantisation costs 3.0 percentage points of accuracy (0.8145 to 0.7844) and 0.013 of QWK (0.9033 to 0.8900), and the FP32 and INT8 models agree on 89.5% of the images. The INT8 logits correlate strongly with the FP32 logits (Pearson r = 0.989; mean error +0.08, standard deviation 0.84). The loss is concentrated in moderate DR, whose recall falls from 0.82 to 0.65, as 33 FP32-moderate images move to mild and 18 to severe (Fig. 5.11), while the recall of the other grades is unchanged or slightly higher. The OCS stays at 0.798, so the ordinal structure is preserved. Even so, the drop for moderate DR is the main cost of the INT8 model, and a larger or class-balanced calibration set and quantisation-aware training are the first remedies to try.

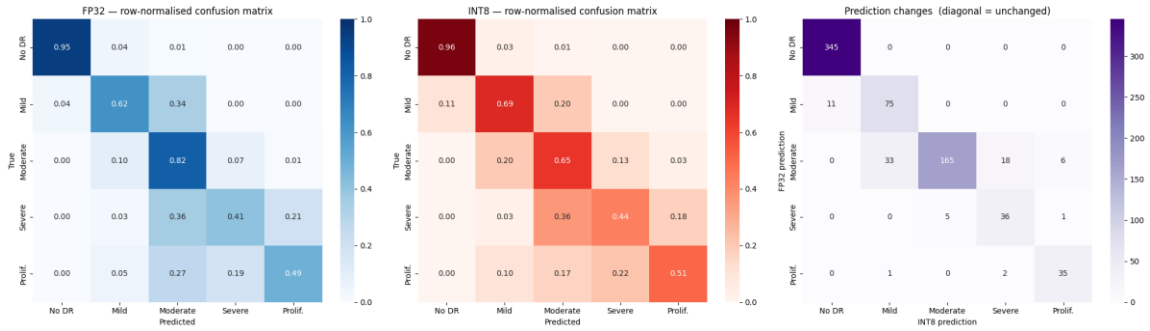


Fig. 5.11. Row-normalised confusion matrices of (a) the FP32 and (b) the INT8 student, and (c) the number of predictions changed by quantisation.

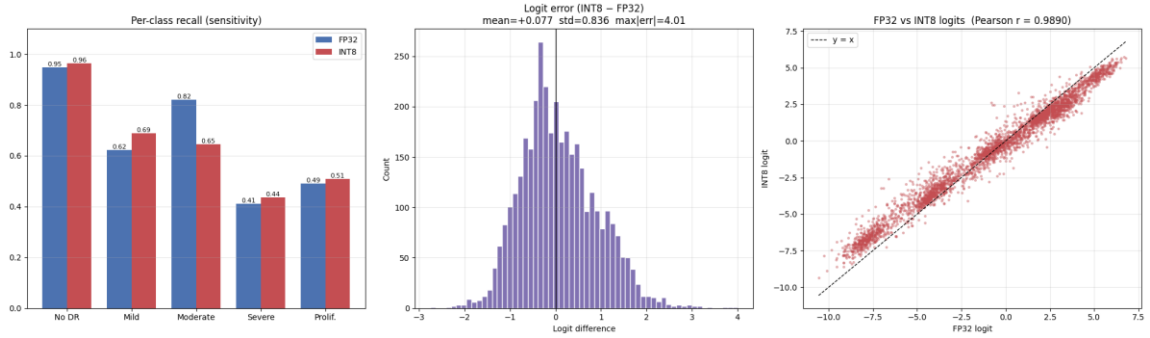


Fig. 5.12. (a) Per-grade recall of the FP32 and INT8 student, (b) the INT8 − FP32 logit error and (c) FP32 against INT8 logits.

5.4.2 Latency

On the x86 Colab CPU the INT8 model is slower than the FP32 model (10.3 against 6.8 ms/image with one thread, and 27.3 against 18.6 ms with four threads), because this CPU has no fast INT8 path for the convolution kernels used. The ranking on the Raspberry Pi 4 may differ, since its ARM Cortex-A72 cores have a different INT8 kernel path in ONNX Runtime. Both models must therefore be timed on the Raspberry Pi itself, and the faster one that keeps acceptable accuracy should be shipped. The FP32 model runs at about 7 ms per image on a single x86 core, but no Raspberry Pi measurement has been made yet, so the Pi rows of Table 5.6 are left as placeholders.

5.4.3 Summary of Differences among the Three Models

Table 5.7 summarises the three models side by side and the difference between each pair. Distillation costs very little (a loss of 0.007 in accuracy and 0.001 in QWK), whereas INT8

quantisation costs about four times more in accuracy and over ten times more in QWK, in return for a file that is 3.4 times smaller.

Table 5.7. Differences in metrics among the teacher, the student (FP32) and the student (INT8).

Metric	Teacher	Student FP32	Student INT8	Δ student − teacher	Δ INT8 − FP32	Δ INT8 − teacher
Parameters (M)	87.54	3.05	3.05	−84.49 (28.7× fewer)	0	−84.49
ONNX size (MB)	373.34	12.21	3.55	−361.13 (30.6× smaller)	−8.66 (3.4× smaller)	−369.79 (105× smaller)
Accuracy	0.8213	0.8145	0.7844	−0.0068	−0.0301	−0.0369
Macro F1-score	0.6820	0.6628	0.6355	−0.0192	−0.0273	−0.0465
QWK	0.9043	0.9033	0.8900	−0.0010	−0.0133	−0.0143
OCS	0.7954	0.7950	0.7981	−0.0004	+0.0031	+0.0027
Latency, x86 1 thread (ms)	–	6.8	10.3	–	+3.5	–

The teacher's OCS and the student's OCS differ by less than 0.001, and the OCS of the INT8 model is slightly higher than both. These differences are well below the run-to-run noise and should not be read as one model being more ordinal than another. The relative loss in QWK is 0.1% from teacher to student and 1.5% from FP32 to INT8.

5.5 Effect of Preprocessing

A controlled ablation of the preprocessing stages has not been run, so only a qualitative assessment is given. Fig. 4.2 shows that the Stage 1 transform removes the large-scale colour and brightness differences between images and makes vessels and lesions stand out against a uniform grey background, for example the dense hard exudates of the severe example. The faint lesions of the mild and proliferative examples remain low in contrast, which is consistent with the lower recall for these grades. The ROI crop removes the black border, so that more of the 384 × 384 input is retina. A quantitative comparison of the input pipelines (no preprocessing, ROI only, Stage 1 only, both) is listed as future work.

5.6 Preliminary Experiment on EyePACS

To check generalisation beyond APTOS, a teacher with the same architecture was trained on the balanced EyePACS pool of 5,000 images (4,000 for training and 1,000 for validation, 200 per grade in the validation set) in a Kaggle notebook. The run was still in progress when this report was written, and distillation on EyePACS has not been done. Table 5.8 shows the validation results after 17 epochs. Validation QWK rose from 0.67 to about 0.85, with a best saved checkpoint of QWK 0.8497 and accuracy 0.6130 at epoch 13, and then

dropped to 0.81 at epoch 17. Accuracy is much lower than on APTOS because the validation set is balanced, so the hard grades (1, 2 and 3) count as much as grade 0, and because EyePACS images are of more variable quality. These numbers are therefore not comparable with the APTOS results and are given only as preliminary evidence. [Update with the final EyePACS results when the run is complete.]

Table 5.8. Preliminary EyePACS teacher run (balanced pool, 1,000 validation images; training in progress).

Epoch	Val. accuracy	Val. QWK	Val. F1	Train QWK	Note
1	0.4260	0.6707	0.3588	0.3878	Backbone frozen
5	0.4750	0.6939	0.4691	0.5388	
8	0.5700	0.8149	0.5722	0.6795	Saved
10	0.5990	0.8419	0.5903	0.7040	Saved
13	0.6130	0.8497	0.5947	0.7135	**Best, saved**
15	0.6230	0.8503	0.6240	0.7486	$\Delta < 0.005$
17	0.5820	0.8110	0.5766	0.7637	Latest logged

5.7 Discussion

5.7.1 Key Findings

- **The compact student matches the teacher in QWK and OCS.** A 3.05 M-parameter MobileNetV3 student reaches QWK 0.903 against 0.904 for the 87.5 M-parameter Swin teacher, and 99% of its accuracy; their OCS values are 0.795 and 0.795. The quantised model is about 105 times smaller than the teacher's ONNX file.
- **Errors are mostly small.** About four of five teacher errors are between neighbouring grades and none confuse the two extremes, so the model is usable for screening: it flags 98.7% of DR cases and 93.6% of referable cases while keeping specificity at 94.5% (student: 99.2%, 91.9% and 94.7%).
- **Why the student can match the teacher.** Distillation is a regulariser. The soft labels carry information about how similar the grades are, and the feature-mimicking term gives the student a richer target than a one-hot label. The teacher has 87.5 M parameters trained on only 2,929 images and over-fits after about 9 epochs, so its extra capacity is not fully used. The validation set (733 images) is also small, so differences of 0.001–0.007 are within noise. Distilled students that equal or slightly exceed their teachers are well known; here the student is slightly below the teacher on every overall metric, which is the usual result.

- **INT8 quantisation has a measurable cost.** Accuracy falls by 3.0 points and QWK by 0.013, mostly through moderate DR, which is the price of a 3.55 MB model; FP32 (12.2 MB) is still small enough for the Pi if the INT8 loss is not acceptable.
- **Rare grades are the weak point.** Severe DR (F1 0.42) and proliferative DR (recall 0.54) are the hardest, because they have the fewest examples (154 and 236 training images). The weighted sampler repeats these images 9.4 and 6.1 times per epoch but cannot create new information.
- **Over-fitting appears early.** The teacher over-fits after about 9 epochs. Using strong augmentation, early stopping and checkpoint selection on QWK is therefore important.

5.7.2 Comparison with Related Work

Published distillation studies report higher accuracies, for example 99.04% on APTOS [19] and 95.39% for DR severity [20]. These figures are not directly comparable with ours: they differ in the splits, in the task definition and in the protocol, and they were obtained on GPU hardware rather than on a Raspberry Pi. Our five-grade accuracy of 82.1% on a held-out stratified validation split, with a QWK above 0.90, is in line with five-class grading on APTOS under a strict split. The contribution of this work lies in the complete, offline system rather than in a higher benchmark number: a stable imaging assembly, a model small enough for a Raspberry Pi, and an evaluation of ordinal behaviour.

5.7.3 Limitations

- **No separate test set.** The validation split is used both to choose the checkpoint and to report results, which gives slightly optimistic figures. A held-out test split, or cross-validation, is needed for the final numbers.
- **Single run, single dataset.** All main results come from one training run on APTOS. Differences of about 0.006 in QWK between epochs are within the run-to-run noise, so small differences between models should not be over-interpreted. External validation on EyePACS and other sources is still pending.
- **No on-device measurement yet.** Latency, memory and power on the Raspberry Pi 4 have not been measured, and the x86 timings do not predict the Pi ranking of FP32 and INT8.
- **Domain shift.** The models were trained on images from commercial fundus cameras. Images from the 20D-lens Raspberry Pi imager differ in field of view, illumination and sharpness, and have so far been tested only qualitatively.

- **No clinical validation.** The system is a screening aid, and no prospective clinical study has been carried out.

Chapter 6

Conclusion and Future Scope

6.1 Summary of the Work

This project designed OpthaEdge-DED, a portable and offline system for grading diabetic retinopathy. The hardware combines a Raspberry Pi 4 Model B, a 20D fundus lens and a focus-oriented stability stand, which removes the examiner-dependent variability of handheld capture. For the software, a high-capacity Swin-Base + CBAM teacher was trained on APTOS 2019 and its knowledge was distilled into a 3.05 M-parameter MobileNetV3-Large student with a lightweight CBAM, using a combined loss of temperature-scaled KL divergence, cross-entropy and feature mimicking. The student is exported to ONNX with its normalisation built in and quantised to INT8 for the Raspberry Pi. The Ordinal Consistency Score (OCS) was defined to test whether ordinal structure survives distillation.

6.2 Conclusions

1. The Swin-Base + CBAM teacher reached an accuracy of 82.13% and a QWK of 0.9043 on the held-out validation split of APTOS 2019, with 87.5 M parameters.
2. The distilled MobileNetV3-Large + lightweight-CBAM student reached an accuracy of 81.45% and a QWK of 0.9033 with 3.05 M parameters, which is 28.7 times fewer, and retained 99.9% of the teacher's QWK, with a macro AUC of 0.942.
3. About 79% of the teacher's errors are between neighbouring grades and none confuse no DR with proliferative DR. Read as a screening test, the model detects 98.7% of DR cases and 93.6% of referable cases, with a specificity of 94.5%.
4. Severe and proliferative DR, the rarest grades, are the weakest (F1 0.42 and 0.62), so more data or targeted methods are needed for them.
5. The student's ONNX file is 12.21 MB in FP32 and 3.55 MB in INT8, against 373 MB for the teacher. INT8 quantisation lowers accuracy to 78.4% and QWK to 0.890, mainly through moderate DR, so the FP32 and INT8 models must be compared on the Raspberry Pi before one is chosen.
6. The OCS of the student (0.7950) matches that of the teacher (0.7954), showing that the ordinal structure was transferred. The web platform (Section 4.10) provides the

clinical workflow around the model; the on-device measurements and the final EyePACS results are the main items still to be completed.

6.3 Weaknesses of the Present Work

The results are encouraging, but the work has weaknesses that a reader should keep in mind (Table 6.1). Some are limits of the data and the experimental design, and some are limits of the models themselves.

Table 6.1. Weaknesses of the present work and the evidence for each.

Weakness	Evidence	Effect
Weak on rare grades	Student F1: severe 0.395, proliferative 0.598, mild 0.575; teacher severe F1 0.42. Only 154 and 236 training images for grades 3 and 4.	Advanced disease is the most urgent to catch, yet is detected least reliably.
Mild ↔ moderate and moderate ↔ severe confusion	34% of mild images and 36% of severe images are predicted as moderate.	Wrong referral priority for borderline cases.
INT8 accuracy loss	Accuracy −3.0 points, QWK −0.013; moderate recall 0.82 → 0.65; 10.5% of predictions change.	The smallest model is the least accurate one.
No separate test set	The same 733 images select the checkpoint and report the results.	Reported metrics are slightly optimistic.
Single run and single dataset	One seed; QWK varies by about ±0.006 between epochs; APTOS only.	Small differences cannot be trusted; generalisation is not shown.
Teacher over-fits	Validation loss rises after epoch 7 while training loss keeps falling.	Part of the teacher's capacity is wasted, and the student inherits its errors.
Domain shift to the prototype	The models are trained on commercial-camera images and have not been tested on Pi-imager images.	Real-world accuracy may be lower than reported.
No on-device measurements	Latency and memory on the Raspberry Pi 4 are not measured; INT8 was slower than FP32 on x86.	The claim of real-time offline use is not yet proven.
No explainability	Grad-CAM++ is not implemented.	Clinicians cannot see why a grade was assigned.
No clinical validation	The platform runs in research-only mode and the models are not approved for clinical use.	Cannot be used for diagnosis.
Preprocessing not ablated	Only a qualitative assessment of the preprocessing stages was made.	The benefit of each stage is not quantified.

6.4 Future Improvements and Scope

- **Complete the evaluation:** measure the latency, memory use and power of the FP32 and INT8 models on the Raspberry Pi 4, and recover the INT8 accuracy with a larger calibration set or quantisation-aware training.

- **Ordinal-aware teacher:** add a differentiable QWK term to the teacher's loss and compare the OCS and QWK of students distilled from a focal-only teacher and a QWK-aware teacher.
- **Stronger validation:** hold out a separate test split or use cross-validation, repeat runs with different seeds, and test on EyePACS and on images captured with the prototype.
- **Preprocessing ablation:** compare no preprocessing, ROI only, Stage 1 only and both, to quantify the effect of each stage.
- **Clinical deployment of the platform:** obtain clinical validation, a production identity provider, encrypted storage, an image-quality model and regulatory approval, so that the platform can leave research-only mode.
- **Explainability:** add Grad-CAM++ heat maps so that clinicians can see which retinal regions drive a prediction.
- **Wider scope:** extend the system to other retinal diseases such as glaucoma and AMD, and to multimodal inputs, and carry out clinical validation in screening camps.

Table 6.2 links each weakness to the improvement planned for it and the benefit expected.

Table 6.2. Planned improvements and the weaknesses they address.

Improvement	How	Addresses
Class-balanced and ordinal training	Add a differentiable QWK term, class-balanced or LDAM loss, mixup/CutMix for the rare grades, and extra data from EyePACS, IDRiD and Messidor-2.	Weak rare grades; mild ↔ moderate confusion
Better INT8 model	Quantisation-aware training, a larger calibration set weighted to moderate DR, mixed precision (FC layer in FP32), and the calibration sweep of notebook section 21.	INT8 accuracy loss
Rigorous evaluation	Separate test split or 5-fold cross-validation, several seeds with confidence intervals, and external validation on EyePACS and prototype images.	No test set; single run; domain shift
Pi benchmarking	Measure latency, peak memory, temperature and power of FP32 and INT8 on the Raspberry Pi 4 and choose the better one; try ONNX Runtime thread and graph-optimisation settings.	No on-device measurements
Regularisation and stronger teacher	Stronger augmentation, stochastic depth, layer-wise learning-rate decay, and multi-teacher or self-distillation.	Teacher over-fitting
Explainability	Grad-CAM++ heat maps and lesion overlays in the web platform; uncertainty and an "ungradable" output for low-quality images.	No explainability; trust

Improvement	How	Addresses
Preprocessing study	Compare no preprocessing, ROI only, Stage 1 only and both; test learned image enhancement.	Preprocessing not ablated
Clinical readiness	Prospective study in screening camps with ophthalmologist grading, a production identity provider, encrypted storage, and regulatory approval.	No clinical validation

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